

Structured English

1. User Authentication

```
BEGIN
DISPLAY login form.
INPUT username and password.
IF username and password are valid THEN
    ALLOW teacher to access the system
ELSE
    DISPLAY "Invalid login credentials"
    RETURN to login page
END IF
END
```

2. Live Camera Monitoring

```
BEGIN
START live camera
CAPTURE video frames continuously
FOR each captured frame DO
    DETECT student using Bounding Box Tracking.
    EXTRACT body keypoints using Pose Estimation.
    STORE captured pose data
END FOR
END
```

3. Pose Validation

```
BEGIN
READ captured pose data
IF pose estimation is complete and valid THEN
    SAVE validated pose estimation
ELSE
    DISCARD invalid pose
END IF
END
```

4. Suspicious Behavior Detection

```
BEGIN
READ validated pose estimation
EXTRACT behavior features
SEND feature set to trained Deep Learning model
IF predicted behavior is suspicious THEN
    FLAG suspicious behavior
    ANNOTATE student pose
    STORE processed behavior
ELSE
    MARK behavior as normal
```

```
END IF
END
```

5. Behavior Classification

```
BEGIN
IF behavior is Side Glancing THEN RECORD Side Glancing
ELSE IF behavior is Hand Signals THEN RECORD Hand Signals
ELSE IF behavior is Standing Up THEN RECORD Standing Up
ELSE IF behavior is Passing Notes THEN RECORD Passing Notes
ELSE IF behavior is Head Down THEN RECORD Head Down
ELSE IF behavior is Bounding Box Collision THEN RECORD Bounding Box
Collision
ELSE RECORD Normal Behavior
END IF
END
```

6. Alert Generation

```
BEGIN
IF suspicious behavior is detected THEN
    GENERATE alert
    DISPLAY flagged behavior to the teacher
    SAVE incident into database
ELSE
    CONTINUE monitoring
END IF
END
```

7. Module Management

```
BEGIN
The teacher selects Manage Modules
IF teacher adds a module THEN SAVE module
ELSE IF teacher edits a module THEN UPDATE module
ELSE IF teacher deletes a module THEN REMOVE module
END IF
END
```

8. Data Retrieval

```
BEGIN
Teacher selects Check P1 Data or Check P2 Data
RETRIEVE corresponding examination records
DISPLAY examination results
END
```

9. Overall System Process

```
BEGIN
Teacher logs into the system
VERIFY login credentials
IF login is successful THEN
    START live camera monitoring
```

```
CAPTURE student movements
PERFORM pose estimation
VALIDATE pose estimation
EXTRACT behavioral features
CLASSIFY behavior using Deep Learning model
IF suspicious behavior is detected THEN
    FLAG the behavior
    ANNOTATE the student
    DISPLAY alert
    SAVE incident into the database
ELSE
    CONTINUE monitoring
END IF
ELSE
    DISPLAY login error
END IF
END
```